

CONTROL EDGE | EPISODE 4

Actuators and Final Control Elements

How software commands become force, motion, heat and flow

Solenoids, pneumatics, hydraulics, control valves, heating, fail-safe design and diagnostics

Audience	Mechanical, process, manufacturing, instrumentation and automation engineers
Target duration	40-44 minutes
Format	Yael - young mechanical engineer; Amir - senior process and control engineer
Core example	Cooling valve commanded to 80% but failing to deliver cooling
Deliverable	Full script + NotebookLM prompt + checklist + sources

Educational material. It does not replace engineering design, risk assessment, sizing, manufacturer instructions or applicable adopted standards.

1. Pre-production alignment check

- The episode continues directly from Episode 3: from the signal received by the controller to the physical effect produced by the system.
- It is not a product catalogue; each actuator is evaluated by energy, load, dynamics, feedback and failure mode.
- Control valves receive added depth because they join mechanics, fluid mechanics and loop behavior.
- Functional safety is presented as a lifecycle boundary without detailed SIL or PL calculations.
- The ending bridges directly to Episode 5 on motors, VFDs, servos and motion control.

Freshness check

The source map was checked against IEC, ISO and ISA. The episode uses IEC 60534-1:2023, ISO 13849-1:2023 and the new IEC 61514:2026 edition for valve positioners.

2. Episode objectives

- Map the complete actuation chain from software command to physical effect and feedback.
- Distinguish command, actuator state and process effect.
- Compare electrical switching, solenoids, pneumatic, hydraulic, electric and thermal actuators.
- Explain control-valve sizing, characteristic, actuator sizing, positioners and common mechanical nonlinearities.
- Design fail-safe behavior around loss of power, air, communication and stored energy.
- Separate normal control from safety-related final-element functions.
- Use diagnostics, proof testing and maintenance evidence to sustain performance.

3. Timing and segment plan

Time	Segment	Purpose
00:00-03:00	Cold open	Command is not motion; motion is not effect.

Time	Segment	Purpose
03:00-06:30	Actuation chain	Map energy, interface, actuator, load and feedback.
06:30-10:30	Electrical switching	Coils, relays, contactors, SSRs and solenoids.
10:30-15:00	Pneumatics	Force, speed, compressibility and loss of air.
15:00-19:00	Hydraulics	Force density, contamination and stored energy.
19:00-25:30	Control valves	Sizing, characteristics, actuators and positioners.
25:30-29:30	Electric mechanisms	Brakes, clutches, steppers and transmissions.
29:30-33:30	Heating	Power interfaces and independent protection.
33:30-38:00	Fail-safe	Safe state and functional-safety boundaries.
38:00-41:30	Diagnostics	Feedback, testing and maintenance.
41:30-44:00	Resolution	Troubleshoot the cooling valve and bridge to motors.

4. Actuation-chain map

Stage	Engineering decision	Typical failure
1. Required effect	Define flow, force, motion, holding, stopping or heat transfer.	Selecting a device before defining the real outcome.
2. Energy source	Electrical power, compressed air, hydraulic pressure, spring or stored energy.	Loss, contamination, insufficient capacity or hazardous residual energy.
3. Power interface	Relay, contactor, SSR, drive, solenoid pilot, I/P or valve positioner.	Inrush, leakage, welded contacts, transients, wrong pressure or bandwidth.
4. Actuator	Convert energy into linear/rotary motion, force, torque or heat.	Undersizing, stall, friction, seal wear, overheating.
5. Transmission and load	Linkage, gearbox, screw, valve trim, gripper, brake or heater surface.	Backlash, looseness, jam, wear, local overheating.
6. Feedback and validation	Confirm position, pressure, current, speed, force or process effect.	Displaying command as if it were actual state.
7. Safe state and lifecycle	Define response to failures, test, maintain and document.	Uncontrolled restart, trapped energy, untested dormant failure.

5. Actuator decision map

Technology	Typical use	Selection questions
Solenoid / relay output	Fast discrete switching, pilot control, isolation.	Voltage, current, inrush, suppression, leakage, duty, response time.
Pneumatic cylinder	Fast linear motion, clamping, sorting, simple positioning.	Force margin, speed, cushioning, air quality, loss-of-air behavior.
Hydraulic cylinder or motor	High force or torque, controlled load motion.	Pressure, filtration, heat, leakage, hose failure, load holding.
Control valve	Continuous regulation of process flow or pressure.	Cv/Kv, DP, fluid state, cavitation, noise, rangeability, shutoff.
Electric linear/rotary actuator	Remote positioning, low air availability, slow high-torque duty.	Torque profile, speed, duty, gearbox, manual override, fail action.
Brake / clutch / latch	Hold, stop, couple or release mechanical	Service vs holding duty, wear, diagnostics,

Technology	Typical use	Selection questions
	energy.	residual motion.
Heater / power controller	Add controlled thermal power.	Load type, current, switching, heat removal, independent high limit.

Core principle

Command, actual state and process effect must be identified separately in I/O, code and HMI.

6. NotebookLM production prompt

Paste-ready instruction

Create an English-only podcast episode of 40-44 minutes using only this document.

Hosts: Yael, a sharp and curious young mechanical engineer; Amir, a calm senior process and control engineer. Use approximately 45% Yael and 55% Amir.

Use the cooling valve commanded to 80% without delivering cooling as the narrative thread, and resolve it only at the end. Repeatedly distinguish command, state and effect. Keep a natural technical dialogue rather than a read lecture. Do not read URLs or clauses. Do not invent Cv, torque, SIL, PL or response-time requirements. Present fail-safe as a result of risk assessment, not a universal rule. Emphasize stored energy and maintenance boundaries.

Required structure: opening; actuation chain; switching and solenoids; pneumatics; hydraulics; control valves; electric actuators and brakes; heating; fail-safe; diagnostics; resolution and bridge to motors. Use the full script as the factual framework, allowing only short natural reactions that do not change meaning.

7. Full episode script

00:00-03:00 | Cold open - the valve that obeyed and still failed

Amir: A reactor cooling valve receives a command to open to eighty percent. The HMI shows eighty percent. The controller output is eighty percent. Yet the reactor temperature keeps rising. Operations conclude that the control loop is badly tuned.

Yael: But the command is not the same as physical motion, and physical motion is not the same as flow.

Amir: Exactly. The positioner may be starved of air, the stem may be sticking, the linkage may be loose, the valve may be undersized, the trim may be damaged, or the differential pressure may be far from the design case. A perfect software command can end at a weak, slow or failed final element.

Yael: So today we follow the opposite direction from Episode 3. The sensor converted physics into information; now the actuator converts information into force, motion, heat or flow.

Amir: And we will keep one rule throughout the episode: command, state and effect are three different things. Good automation must know which of the three it is displaying and protecting.

Yael: We will cover solenoids, pneumatic and hydraulic cylinders, control valves, heaters, brakes, clutches and other loads, then finish with fail-safe design and the cooling-valve fault.

03:00-06:30 | The actuation chain

Yael: What is the complete chain between a PLC bit and a moving mechanism?

Amir: Start with the demanded process or machine effect: stop a shaft, clamp a part, meter steam, lift a load or add heat. The controller calculates a command. An output module or network transmits it. A power interface provides the required voltage, current, pressure or flow. The actuator converts energy into motion or heat. The transmission mechanism applies it to the load. Feedback confirms state, and the process responds.

Yael: So a digital output is rarely driving the real load directly.

Amir: Correct. It may energize an interposing relay, contactor, solid-state relay, solenoid coil, drive enable or valve positioner. The interface must handle inrush current, inductive energy, switching frequency, isolation and fault detection. A relay contact is not an infinite-life logic symbol.

Yael: And every layer has a different failure mode.

Amir: Yes. The PLC can command ON while a fuse is open. The contactor can pull in while one power pole is damaged. A solenoid can energize while the spool is stuck. A cylinder can move while the load is mechanically jammed. Feedback must be chosen around the actual hazard and required effect.

06:30-10:30 | Discrete electrical actuators and switching

Yael: Let us start with simple outputs: coils, relays, contactors and solenoids.

Amir: A coil is an inductive load. When current is interrupted, the magnetic field collapses and generates a voltage transient. Suppression protects the output but changes release time. A diode gives strong suppression on DC coils but can slow dropout; other suppressors trade voltage for speed. That matters when a fast release is part of a safety or cycle-time requirement.

Yael: What must be checked before assigning a PLC output?

Amir: Steady current, inrush, leakage current, minimum load, switching frequency, short-circuit behavior, common grouping, diagnostic coverage and whether the load is AC or DC. Also check whether a de-energized output truly removes energy or only removes a command while stored energy remains.

Yael: Mechanical relays versus solid-state relays?

Amir: Mechanical relays provide galvanic separation and tolerate many load types, but contacts wear, bounce and can weld. Solid-state devices switch quickly and silently, yet have leakage, voltage drop, heat dissipation and characteristic failure modes. For heaters, an SSR may provide frequent proportional time control, while a contactor may serve isolation or backup shutdown. The architecture comes from the risk and duty cycle.

Yael: A solenoid valve is also not just a coil.

Amir: Right. The coil creates magnetic force, but the valve has an armature or pilot stage, seals, orifices and flow limits. Pilot-operated valves may not shift below a minimum differential pressure. Contamination, moisture, exhaust restriction and wrong porting can defeat the command.

10:30-15:00 | Pneumatics - fast, clean and compressible

Yael: Why are pneumatic cylinders so common in machines?

Amir: They are simple, fast, tolerant of overload and easy to distribute across a factory. But air is compressible. Position stiffness, speed stability and force depend on pressure, flow, volume, cushioning, friction and load. A cylinder sized only from bore times supply pressure ignores losses, rod area, breakaway friction, pressure drop and dynamic margin.

Yael: How should speed be controlled?

Amir: Usually by controlling exhaust flow - meter-out - because the compressed air on the exhausting side provides damping. Meter-in can become unstable with overrunning loads. The correct choice depends on

load direction and motion. Cushioning, shock absorbers and mechanical stops must handle kinetic energy; the end sensor should not be used as the physical stop.

Yael: What happens during loss of air?

Amir: That is a design question, not a universal answer. A vertical axis may fall unless mechanically held. A gripper may release or remain trapped by check valves. A directional valve may spring to a defined position, hold the last position or vent. The safe state must consider trapped pressure, gravity, maintenance and restart. Dumping a manifold can itself create hazardous motion.

Yael: Which standards form the baseline?

Amir: ISO 4414 provides general rules and safety requirements for pneumatic systems on machinery. ISO 15552 standardizes important mounting dimensions for a common cylinder series, but dimensional interchangeability does not guarantee equivalent dynamics, seals or lifetime.

15:00-19:00 | Hydraulics - force density and stored energy

Yael: Hydraulics give much higher force density.

Amir: Yes, with relatively incompressible fluid, high pressures and precise force control. But leakage, contamination, temperature, hose failure, injection injury and stored energy become central. Accumulators and suspended loads can remain hazardous after electrical power is removed.

Yael: What is commonly missed in control design?

Amir: Valve overlap, pressure compensation, load-induced motion, cavitation, thermal expansion of trapped fluid and the difference between commanding a directional valve and controlling velocity. Servo and proportional valves need clean fluid, suitable filtration and stable supply. A cylinder may creep because of internal leakage or load, not software.

Yael: How do we stop a vertical hydraulic load safely?

Amir: Often with load-holding valves, counterbalance valves, pilot-operated checks or a mechanical brake, selected and validated for the hazard. Closing a directional valve at the manifold may not prevent hose-failure motion. ISO 4413 establishes general safety requirements, and the machinery risk assessment determines the protective architecture.

Yael: So isolation requires dissipation and verification of pressure, not just an OFF command.

Amir: Exactly. Lockout must address every energy source and stored-energy location.

19:00-25:30 | Control valves - where loop theory meets mechanics

Yael: Let us return to process control valves. What makes them more than a pipe component?

Amir: A control valve is a variable restriction assembled from body, trim, actuator and usually a positioner. Selection combines fluid mechanics, materials, pressure boundary, actuator thrust or torque, dynamic response, leakage requirements and maintainability. IEC 60534 is the central international series for industrial-process control valves; ISA-75 provides closely related standards and practices.

Yael: Why can a valve with the correct line size still control badly?

Amir: Because line size is not valve capacity. The valve is sized for required flow, pressure drop, fluid properties, choked flow, cavitation or flashing risk, noise and operating range. An oversized valve may operate near the seat where small movements create large flow changes, worsening resolution and wear. An undersized valve saturates fully open.

Yael: And inherent characteristic is not installed characteristic.

Amir: Correct. Linear, equal-percentage and quick-opening describe inherent flow behavior under specified test conditions. In the installed system, pump curve and piping losses change the pressure drop across the

valve. Equal-percentage trim is common because it can compensate for changing system gain, but it is not automatically best.

Yael: What does the actuator have to overcome?

Amir: Packing friction, seat load, pressure forces, unbalance, dynamic forces and required shutoff. Pneumatic diaphragm and piston actuators are common; electric and hydraulic actuators are used when their characteristics fit. The actuator must be sized for the worst credible pressure and failure mode, not only normal modulation.

Yael: What does the positioner add?

Amir: It closes a local position loop around the actuator and valve. It improves positioning against friction and pressure forces, converts command to pneumatic output, and modern devices provide diagnostics. But position feedback does not prove flow. A detached stem connection can report actuator travel while the plug does not move as intended.

Yael: Which mechanical problems appear in trends?

Amir: Deadband, hysteresis, stiction, backlash, air-supply limitation and slow response. The loop may oscillate because the valve sticks and releases, not because PID tuning is aggressive. Step tests and signatures can separate controller behavior from final-element dynamics.

25:30-29:30 | Electric actuators, brakes, clutches and mechanical transmission

Yael: Not every controlled movement deserves a full servo system.

Amir: True. Electric linear actuators, geared motors, stepper drives, brakes, clutches and latching mechanisms cover many duties. The selection starts with force or torque versus speed, stroke or angle, duty cycle, backlash, holding requirement, environment and what happens without power.

Yael: A brake is an actuator too.

Amir: Yes. A spring-applied, electrically released brake can hold a vertical axis after power loss, but its stopping capability, wear, air gap, thermal capacity and diagnostic strategy must be engineered. A holding brake is not always rated as a service brake. Likewise a clutch can fail to transmit or fail to release.

Yael: What about stepper motors losing position?

Amir: Open-loop steppers infer position from commanded steps. Excess torque demand, acceleration or resonance can cause missed steps with no automatic awareness. Closed-loop feedback or homing can detect or correct some faults, but mechanical couplings, belts and screws still require inspection. We will treat motors and motion control in a dedicated episode.

29:30-33:30 | Heating and thermal final elements

Yael: Heating has no visible stem, so engineers sometimes treat it as easy.

Amir: Yet heaters combine high energy, thermal lag and fire or product-damage hazards. The controller may switch a contactor, SSR, thyristor controller or power regulator. The load can be resistive, inductive or have temperature-dependent resistance. Wiring, protection and heat removal must be designed for the real current and ambient conditions.

Yael: How should protection be separated from temperature control?

Amir: The normal control loop regulates temperature. An independent high-temperature protective function should detect a dangerous condition and remove or limit energy through an adequately independent path. A welded SSR can remain conducting after the logic output turns off, so a separate contactor or other isolation element may be required by the risk assessment.

Yael: And sensor placement matters again.

Amir: Very much. A heater can overheat locally while the bulk sensor remains below setpoint. Flow proving, low-level protection, surface temperature, overtemperature cutouts and cooldown logic may all be necessary. The final effect is heat transfer, not simply electrical ON.

33:30-38:00 | Fail-safe, de-energize-to-trip and the meaning of safe state

Yael: People often ask whether a valve should fail open or fail closed.

Amir: That question is incomplete. Safe state depends on the hazard and the initiating failure. Cooling water may fail open to protect temperature, but uncontrolled cooling can create another hazard. Fuel gas usually tends toward isolation, but venting, trapped inventory and depressurization must be considered. Some applications require fail-in-place or a controlled sequence rather than one static position.

Yael: How does spring return help?

Amir: It stores mechanical energy to drive the actuator toward a defined position when motive energy is lost. But spring direction, available force, travel time, valve torque profile and common-cause failures must be verified. Stored energy is useful for safety but can also be hazardous during maintenance.

Yael: Is de-energize-to-trip always safer?

Amir: It often reveals loss of power and allows a spring-return final element to move safe, but it is not universal. Spurious trips, energized-to-trip applications, battery-backed systems and continuously powered actuators require analysis. Functional safety standards treat the complete safety function from sensor through logic solver to final element, including proof testing and failure data.

Yael: What is the difference between machine and process frameworks here?

Amir: For machinery, ISO 13849-1 and IEC 62061 provide methods for safety-related control systems across electrical, pneumatic, hydraulic and mechanical technologies. For process SIS, IEC 61511 covers lifecycle requirements. None of them allows us to label a single certified component and declare the full function safe.

38:00-41:30 | Diagnostics, testing and maintenance

Yael: How do we know an actuator will work when demanded?

Amir: Use appropriate feedback, diagnostics and periodic testing. End switches can confirm terminal position; analog feedback can show travel; pressure switches can confirm supply; motor current can reveal load; smart positioners can trend friction and travel deviation. But diagnostics must be independent enough to detect the relevant fault.

Yael: What is partial-stroke testing?

Amir: For some shutdown valves, a small controlled movement is periodically commanded to reveal sticking or actuator faults without fully interrupting the process. It improves diagnostic coverage for certain failures but does not replace full proof testing and cannot reveal every failure. Test intervals and procedures belong to the safety lifecycle.

Yael: Maintenance can change the control response.

Amir: Absolutely. Repacking a valve changes friction; replacing a cylinder changes cushioning; fitting a different solenoid changes flow; changing suppressors changes release time. As-found and as-left evidence, configuration control, spare equivalence and functional testing are part of control engineering.

41:30-44:00 | Resolve the cooling-valve fault and close

Yael: Back to the reactor cooling valve. What is the troubleshooting sequence?

Amir: First compare command, actual position and process effect on the same timeline. Second verify instrument air pressure and flow during travel. Third perform a safe stroke or step test and observe response time, deadband and friction. Fourth inspect linkage, stem, packing and travel calibration. Fifth verify valve

sizing and available pressure drop at the current operating condition. Sixth inspect trim for blockage or damage. Only after that revisit PID tuning.

Yael: Suppose the HMI showed eighty percent because it was displaying command, not feedback.

Amir: Then the screen told the truth about the software and encouraged a false conclusion about the machine. Rename it Output Command, add validated position feedback where justified, and make bad or stale feedback visible.

Yael: The engineering checklist is therefore: required effect, energy source, actuator and transmission, force and speed margin, feedback, failure position, stored energy, manual intervention, environment, maintenance and proof of operation.

Amir: Exactly. The final element is where code becomes consequence. Design it with the same rigor as the algorithm.

Yael: In Episode 5 we move into electric motors, VFDs, servos and motion control - how torque, speed and position become coordinated machine behavior.

Amir: Until then, never confuse a green output bit with a completed physical action.

8. Producer and host notes

- Maintain a consistent distinction between command, feedback and effect.
- Do not turn technologies or vendors into commercial rankings.
- Stop before detailed control-valve sizing and refer to competent engineering and process data.
- Treat loss of power, air, hose integrity, communication and stored energy as separate scenarios.
- Do not describe partial-stroke testing as a replacement for full proof testing.
- Keep the fault resolution for the closing segment.

9. Engineering actuation-chain checklist

1. What physical effect is required in normal, upset and emergency states?
2. What is the energy source and its capacity, quality and isolation method?
3. What force or torque, speed, stroke, duty cycle and margin are required?
4. What inrush, transient, leakage, heat and switching-frequency limits apply?
5. What friction, backlash, compliance, cushioning and kinetic energy exist?
6. Which feedback proves state and which feedback proves effect?
7. What happens on loss of power, air, pressure, network or sensor?
8. What stored energy remains and how is it dissipated and verified?
9. What is the safe state, who defined it and how is it validated?
10. How are manual override, lockout and restart controlled?
11. Which acceptance, stroke, proof and periodic tests are required?
12. How are as-found/as-left evidence, spares and configuration maintained?

10. Glossary

Term	Meaning in this episode
Actuator	Device that converts supplied energy and a command into mechanical motion, force, torque or another physical output.
Final control element	The part of a process control loop that directly changes the manipulated variable, commonly a control valve or drive.
Final element	In functional safety, the subsystem that acts on the process to achieve or maintain a safe state.
Stiction	Static friction behavior that prevents movement until force builds and the

Term	Meaning in this episode
	mechanism breaks free.
Deadband	Range of input change that produces no corresponding output change.
Fail-safe	Designed behavior that moves or maintains the system in a defined safe condition for specified failures.
De-energize-to-trip	Architecture in which removal of energy initiates the protective action.
Proof test	Periodic test intended to reveal dangerous hidden failures not detected automatically.
Partial-stroke test	Limited valve movement used to detect some failures without executing a full shutdown stroke.
Stored energy	Energy retained in springs, pressure, gravity, capacitance, rotating inertia or heat after command or power removal.

11. Standards and source map

Use note

For every project, verify the edition adopted by contract, jurisdiction and sector, including amendments and manufacturer requirements.

13. IEC 60534-1:2023 - Control valve terminology and general considerations. [Official source](#)
14. IEC 60534-2-4:2009 - Inherent flow characteristics and rangeability. [Official source](#)
15. IEC 60534-2-3:2015 - Flow-capacity test procedures for control valves. [Official source](#)
16. IEC 60534-4:2021 - Inspection and routine testing of control valves. [Official source](#)
17. IEC 60534-7:2010 - Control valve data-sheet requirements. [Official source](#)
18. IEC 61514:2026 - Performance evaluation of pneumatic-output valve positioners. [Official source](#)
19. ISA-75 series - Control valve design, sizing, testing, performance and diagnostics standards. [Official source](#)
20. ISO 4414:2010 - General rules and safety requirements for pneumatic fluid power systems. [Official source](#)
21. ISO 4413:2010 - General rules and safety requirements for hydraulic fluid power systems. [Official source](#)
22. ISO 15552:2018 - Interchangeability dimensions for a common pneumatic cylinder series. [Official source](#)
23. ISO 12100:2010 - Machinery risk assessment and risk reduction principles. [Official source](#)
24. ISO 13849-1:2023 - Safety-related parts of machinery control systems, including non-electrical technologies. [Official source](#)
25. IEC 62061:2021+A1:2024 - Functional safety of safety-related machinery control systems. [Official source](#)
26. IEC 60204-1:2016+A1:2021 - Electrical equipment of machines, including power-drive and emergency-stop requirements. [Official source](#)
27. IEC 61511-1:2016+A1:2017 - Safety instrumented systems lifecycle requirements for process industries. [Official source](#)
28. ISA-84 series - Guidance and standards for SIS from sensor through logic solver to final element. [Official source](#)

12. Episode quality gate

- The listener can draw a complete command-to-effect chain.
- Command, state and effect are not treated as synonyms.
- Pneumatics and hydraulics include stored energy and supply-loss behavior.
- Control valves are explained through sizing, dynamics, positioners and failure modes.
- Fail-safe is presented as a result of risk assessment.

- Functional safety stays within correct responsibility boundaries.
- The ending bridges directly to motors and motion control.